

Examiner
only

4. (a) Heat is supplied to an ideal gas in a sealed container of fixed volume. Explain carefully the physical processes that cause an increase in the **temperature and pressure** of the gas. [6 QER]

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- (b) A container of fixed volume contains oxygen gas at a temperature of 293 K.
- (i) Five oxygen molecules have speeds 400, 425, 450, 550 and 625 m s⁻¹. Determine their rms speed. [2]

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- (ii) Using an appropriate calculation, determine whether or not the rms speed calculated in part (b)(i) is consistent with the expected rms speed of the molecules of the gas at this temperature. (Relative molecular mass of oxygen gas = 32.) [3]

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- (iii) If the gas in the container is heated and the pressure of the gas increases by 20% of its initial value, determine the rms speed of the molecules of the gas. [2]

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